Elite Online Learning Platform

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MINI LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE414: Mobile Application Design Lab in the Computer Science and Engineering Department**

DAFFODIL INTERNATIONAL UNIVERSITY

Dhaka, Bangladesh

April 22, 2026

DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Name of the course teacher**, **course teacher's Designation**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

Submitted To:

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Lecturer
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COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:.

Table 1: Course Outcome Statements

CO's	Statements
CO1	**Define** and **Relate** classes, objects, members of the class, and relationships among them needed for solving specific problems
CO2	**Formulate** knowledge of object-oriented programming and Java in problem solving
CO3	**Analyze** Unified Modeling Language (UML) models to **Present** a specific problem
CO4	**Develop** solutions for real-world complex problems **applying** OOP concepts while evaluating their effectiveness based on industry standards.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP
CO1	PO1	C1, C2	KP3	EP1, EP3
CO2	PO2	C2	KP3	EP1, EP3
CO3	PO3	C4, A1	KP3	EP1, EP2
CO4	PO3	C3, C6, A3, P3	KP4	EP1, EP3

The mapping justification of this table is provided in section **4.3.1**, **4.3.2** and **4.3.3**.

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Chapter 1

Introduction

Every chapter should start with 1-2 sentences on the outline of the chapter.

1.1 Introduction

This report presents the development of the Elite App, a web-based skill learning platform designed to provide high-quality educational content through verified expert instructors. The system enables learners to access structured courses, participate in live interactive sessions and engage in collaborative discussions to enhance their learning experience. This project addresses common challenges in existing online learning platforms, such as unverified content, lack of structured learning paths, and limited real-time interaction between instructors and students. By introducing a controlled content environment and interactive learning features, the Elite App aims to deliver a more reliable, engaging, and effective digital education system.

1.2 Motivation

The development of the Elite App is driven by the growing need for reliable and high-quality digital learning platforms in today's technology-driven world. As online education continues to expand, many existing platforms face challenges such as unverified content, lack of structured learning, and limited interaction between learners and instructors. This project aims to address these issues by creating a controlled and expert-driven learning environment. By focusing on verified instructors and interactive learning features, the Elite platform aligns with modern educational demands and provides a scalable solution for effective skill-based learning.

1.3 Objectives

1. To design and develop a user-friendly platform that enables efficient course creation, management and access for learners.
2. To implement a verified instructor system that ensures only qualified experts can create and publish educational content.
3. To integrate role-based access control for administrators, instructors, and learners to manage system functionality effectively.
4. To provide interactive learning features such as live sessions, discussions, and chat systems to enhance user engagement.
5. To ensure secure user authentication and data management using modern security practices.
6. To enable structured learning through organized courses and track user participation and progress within the platform.

1.4 Feasibility Study

Existing online learning platforms provide a wide range of educational services; however, many of them either lack proper content quality control or are overly complex and costly for users. Platforms that allow open content creation often suffer from inconsistent quality, while premium platforms may limit accessibility due to subscription costs. The development of the Elite App offers a feasible solution by combining high-quality, expert-verified content with a simplified and user-friendly system. By focusing on essential features such as structured courses, live interaction, and role-based access, the platform ensures both functionality and accessibility while reducing unnecessary complexity.

1.5 Gap Analysis

The primary gap identified in current e-learning systems is the absence of a controlled and quality-focused learning environment. Many platforms do not verify instructors, which leads to unreliable educational content and unstructured learning experiences. Additionally, there is limited integration of real-time interaction and collaborative learning tools in a single system. The Elite App addresses these gaps by introducing a verified instructor model, structured course organization, and interactive features such as live sessions and discussion systems. This approach ensures a more reliable, engaging, and effective learning platform tailored to modern educational needs.

1.6 Project Outcome

Upon completion, the Elite App delivers a fully functional skill-based learning platform that enables verified instructors to create and manage structured courses while allowing learners to enroll, access educational content, and participate in interactive learning activities. The system includes key features such as course creation and enrollment, video-based learning, live interactive sessions, real-time communication through chat and discussions, and an administrative panel for managing users and content. Overall, the project provides a reliable, user-friendly, and quality-focused digital learning environment that enhances both teaching and learning experiences.

Chapter 2

Proposed Methodology/Architecture

Every chapter should start with 1-2 sentences on the outline of the chapter.

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

The system follows a three-tier architecture:

1. **Frontend:** HTML templates rendered by Flask.
2. **Backend:** Flask framework for routing and business logic.
3. **Database:** SQLite for storing user and course data.

2.1.2 Proposed Methodology/ System Design

1. **Login and Registration Module:** User roles include student, instructor, and admin.

2. **Course Module:** Instructors can create courses with multimedia content.

3. **Enrollment System:** Students enroll in courses and access materials. .

4. **Admin Dashboard:** Centralized control for user and course management.

Figure 2.1: This is a sample diagram

2.1.3 UI Design

Home

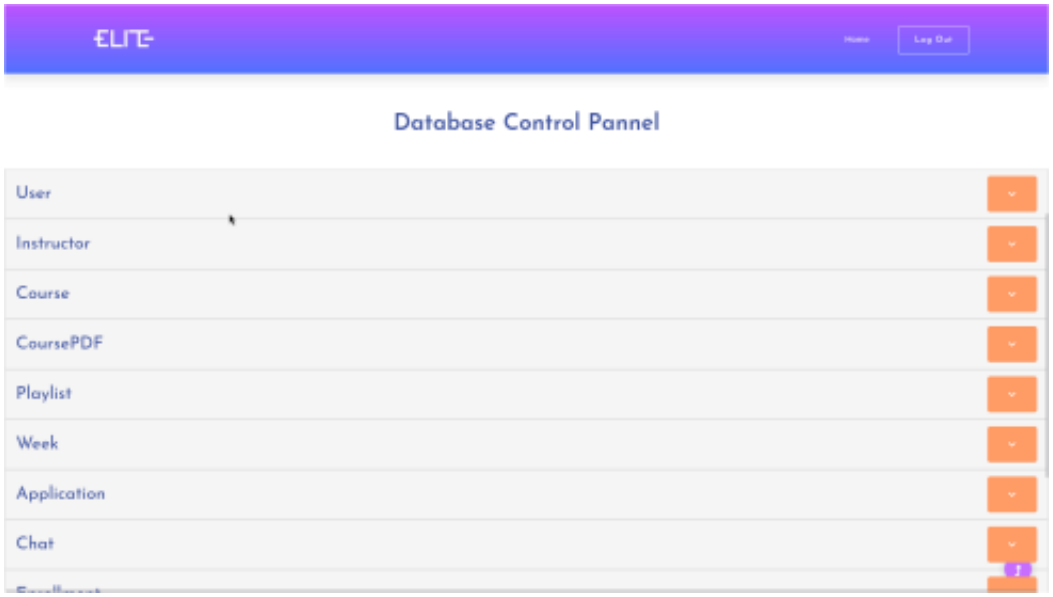
Log - in

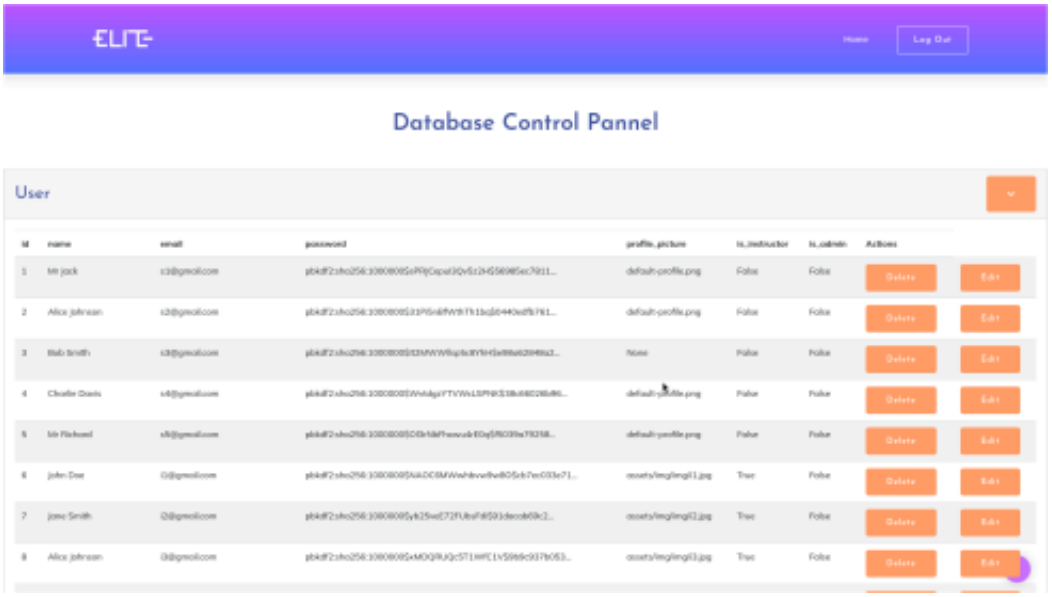
Registration

Admin

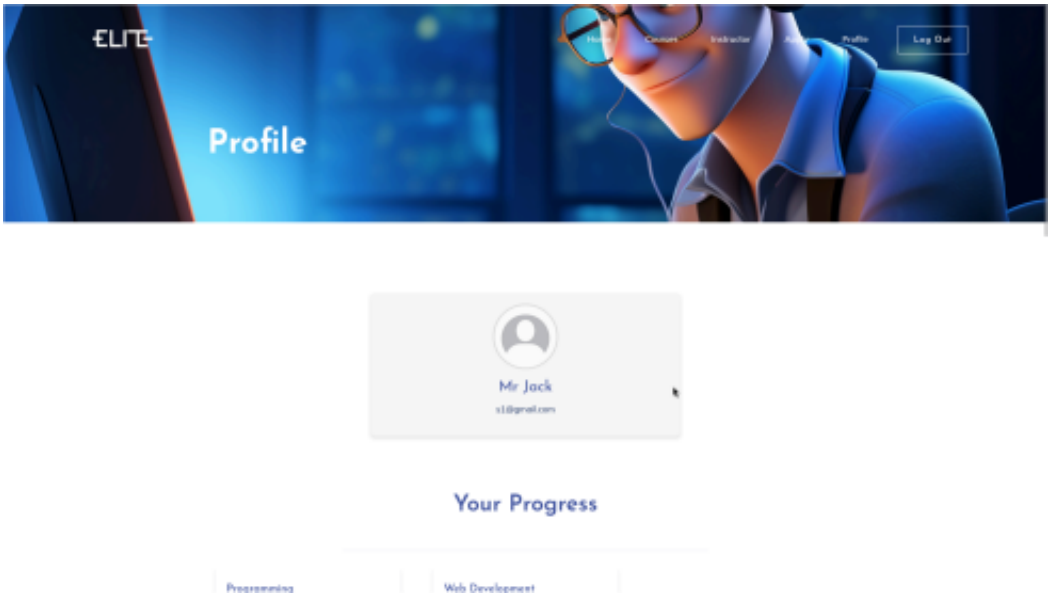
Applications

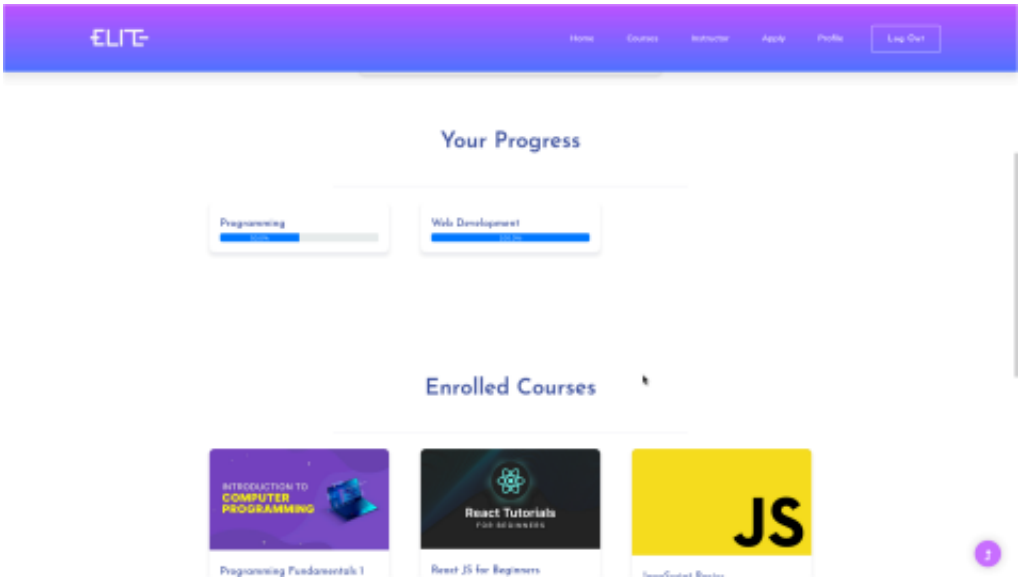
Email	Name	Actions
1@1.com	1 1	Accept




ELITE
Home
Log Out
Database Control Pannel
User

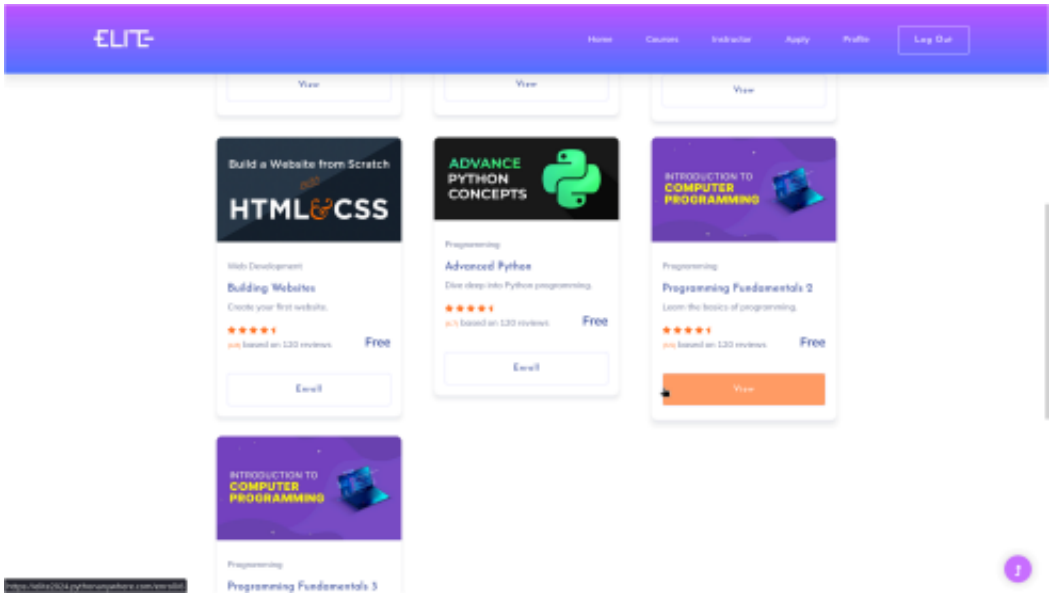
Profile


ELITE
Profile
Log Out
Mr Jack
Your Progress
Programming
Web Development

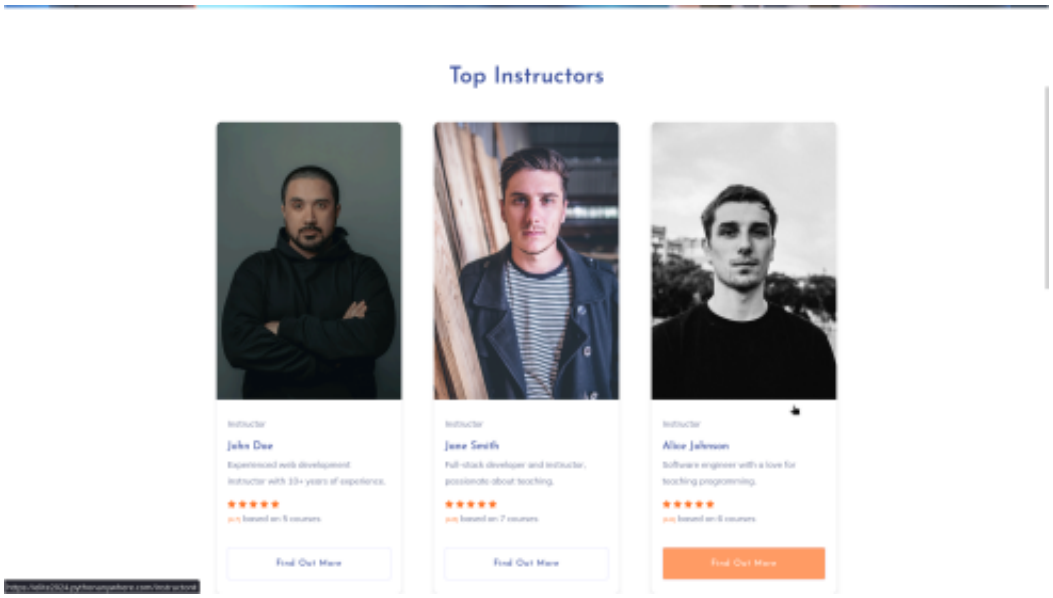

ELITE
Your Progress
Enrolled Courses
INTRODUCTION TO COMPUTER PROGRAMMING
React Tutorials
JS

ELITE
Enrolled Courses
INTRODUCTION TO COMPUTER PROGRAMMING
Programming Fundamentals 1
View
React Tutorials
FOR BEGINNERS
React JS for Beginners
View
JS
JavaScript Basics
View
INTRODUCTION TO COMPUTER PROGRAMMING
Programming Fundamentals 2
View

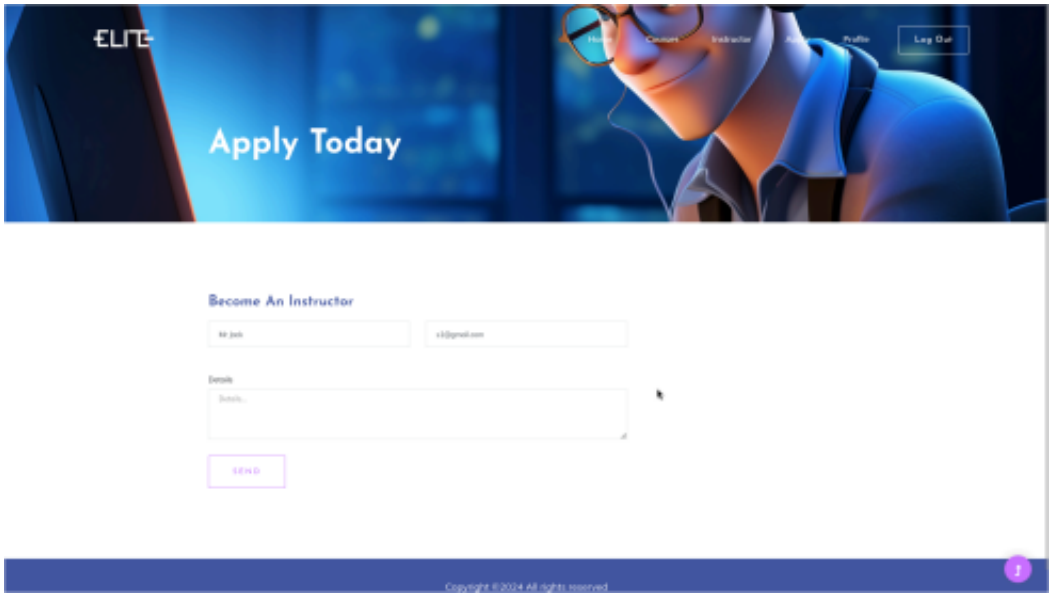
Course



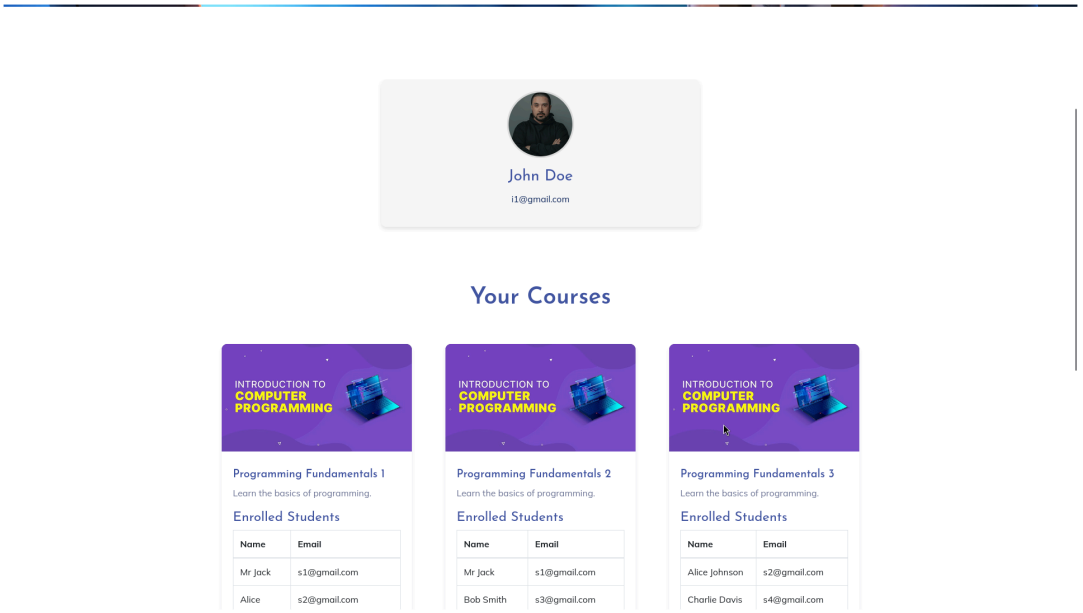
Instructor

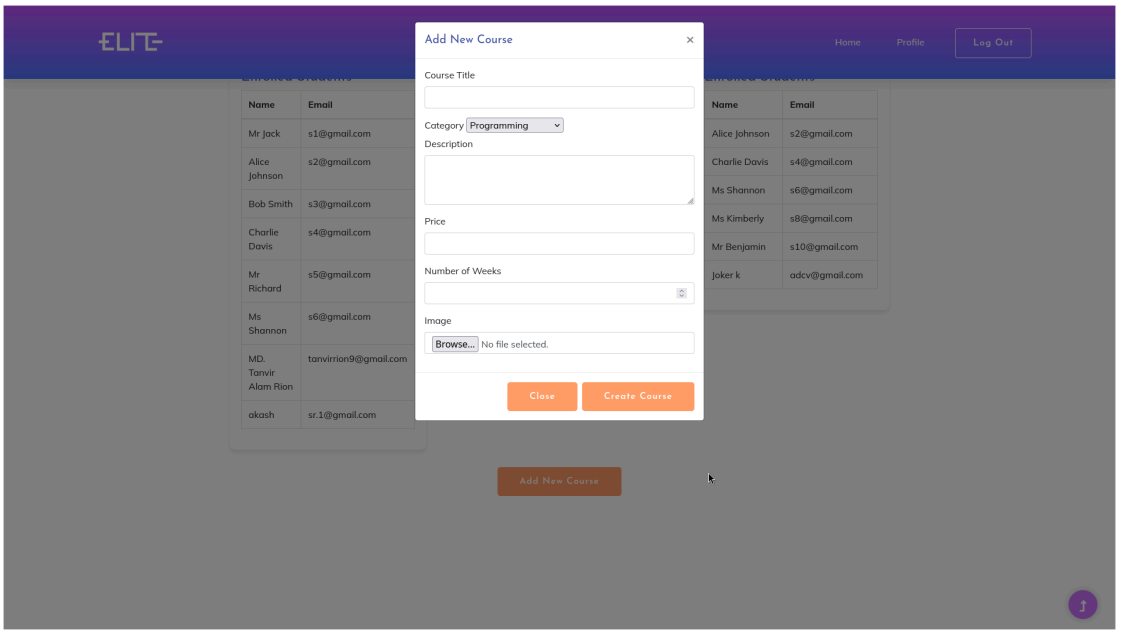


Apply

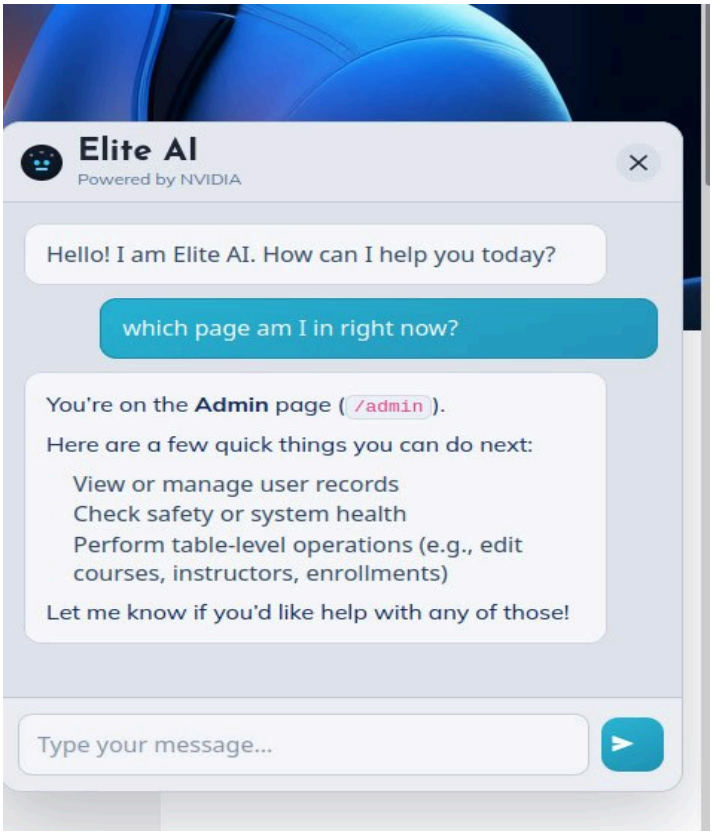


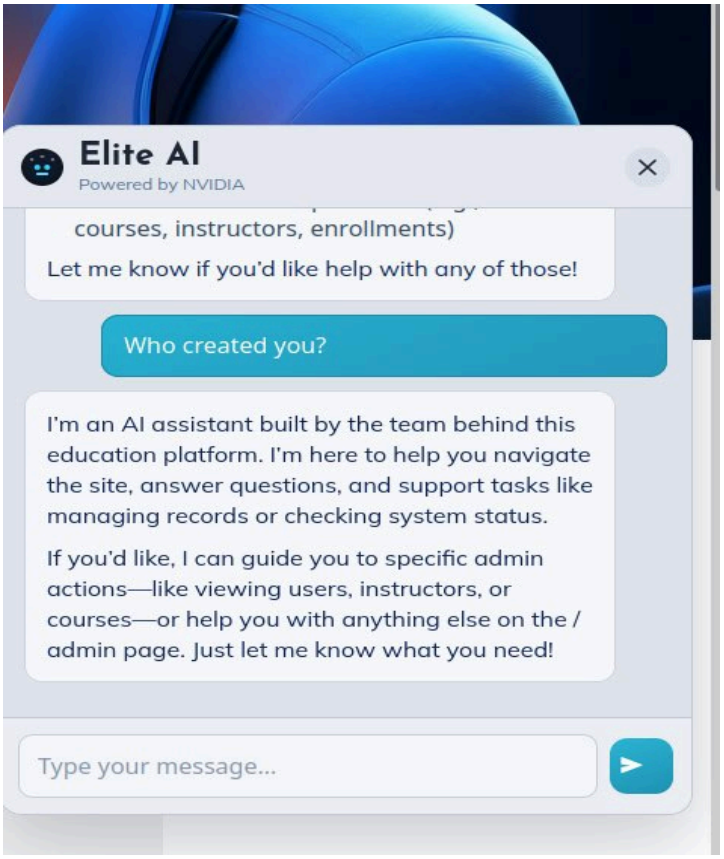
Instructor Profile




ELITE
Home
Profile
Log Out
Add New Course
Course Title
Category
Programming
Description
Price
Number of Weeks
Image
Browse...
No file selected.
Close
Create Course
Name
Email
Mr Jack
s1@gmail.com
Alice Johnson
s2@gmail.com
Bob Smith
s3@gmail.com
Charlie Davis
s4@gmail.com
Mr Richard
s5@gmail.com
Ms Shannon
s6@gmail.com
MD. Tanvir Alam Rion
tanvirrion9@gmail.com
akash
sr.1@gmail.com
Alice Johnson
s2@gmail.com
Charlie Davis
s4@gmail.com
Ms Shannon
s6@gmail.com
Ms Kimberly
s8@gmail.com
Mr Benjamin
s10@gmail.com
Joker k
adcv@gmail.com
Add New Course

AI Chatbot


Elite AI
Powered by NVIDIA
Hello! I am Elite AI. How can I help you today?
which page am I in right now?
You're on the Admin page (/admin).
Here are a few quick things you can do next:
View or manage user records
Check safety or system health
Perform table-level operations (e.g., edit courses, instructors, enrollments)
Let me know if you'd like help with any of those!
Type your message...


Elite AI
Powered by NVIDIA
courses, instructors, enrollments)
Let me know if you'd like help with any of those!
Who created you?
I'm an AI assistant built by the team behind this education platform. I'm here to help you navigate the site, answer questions, and support tasks like managing records or checking system status.
If you'd like, I can guide you to specific admin actions—like viewing users, instructors, or courses—or help you with anything else on the /admin page. Just let me know what you need!
Type your message...

2.2 Overall Project Plan

The Elite Online Learning Platform is a mobile application designed to provide an easy and interactive learning experience for students. The app will include features such as user registration and login, course browsing, video-based lessons, quizzes and progress tracking.

The project will be developed using a mobile framework like Flutter or native Android, with a backend service such as Firebase for authentication, database, and storage. The system will follow a simple architecture where the mobile app communicates with the backend to manage users and course content.

The development process will be completed in several phases, including planning, UI design, implementation of core features, testing, and final deployment. The goal is to build a user-friendly and functional learning platform within a limited time suitable for a project.

Overall, this project will demonstrate practical skills in mobile app development, UI/UX design, and basic backend integration.

Target Users

1. Students (school, college, university)
2. Self-learners
3. Beginners in programming/design/business

Core Features

User Module

1. Registration & Login
2. Profile management
3. Progress tracking

Course Module

1. Course list (categories: Programming, Design, Business)
2. Course details (title, instructor, duration)
3. Video lessons

Learning Module

1. Watch lessons
2. Mark lessons as completed
3. Resume learning

Quiz Module

1. MCQ-based quizzes
2. Instant results
3. Score tracking

Search & Filter

1. Search courses
2. Filter by category

Chapter 3

Implementation and Results

Every chapter should start with 1-2 sentences on the outline of the chapter.

3.1 Implementation

Development Environment

The Elite platform was developed using a modern web-based development environment that supports both frontend and backend functionalities. The system was built using the following technologies:

1. **Programming Language:** JavaScript and Python
2. **Frontend:** HTML, CSS, Bootstrap, and JavaScript
3. **Backend Framework:** Node.js / Django (depending on implementation)
4. **Database:** MySQL / MongoDB for structured data storage
5. **Tools:** VS Code for development, Postman for API testing, and GitHub for version control

The chosen technologies ensure flexibility, scalability, and efficient handling of dynamic user interactions. The use of modern frameworks allows rapid development and smooth system performance.

Key Functional Modules

Authentication Module:
The authentication system allows users to register and log in securely with role-based access (administrator, instructor, and learner). User credentials are protected using secure encryption and authentication mechanisms to ensure data safety.

Course Management Module:
Instructors can create, update, and manage structured courses. Each course includes essential information such as title, description, video lectures, and additional learning materials.

Enrollment System:
Learners can browse available courses and enroll based on their interests. The system maintains relationships between users and courses to track enrollment effectively.

Live Session and Interaction Module:
The platform supports live interactive sessions where instructors can conduct real-time classes. Chat and discussion features enable direct communication between learners and instructors.

Progress Tracking System:
The system tracks user activity and course participation, allowing learners to monitor their progress throughout the course.

Admin Dashboard:
A centralized admin panel allows administrators to manage users, verify instructors, monitor content, and maintain system quality.

Database Design

The Elite platform uses a structured database system to manage application data efficiently. The main database tables include:

Users: Stores user information such as ID, name, role, and authentication details
Courses: Contains course-related data including title, description, and instructor information
Enrollments: Maintains relationships between users and enrolled courses
Content: Stores course materials such as videos and articles
Progress: Tracks learner activity and course completion status

This design ensures efficient data management and supports the overall functionality of the system.

3.2 Performance Analysis

Scalability Testing

The system was tested under simulated user loads to evaluate performance. The Elite platform demonstrated stable performance with multiple concurrent users due to its efficient backend design and lightweight architecture.

Query Optimization

Database queries were optimized using indexing and efficient data retrieval methods. This improves system responsiveness, especially when handling large numbers of users and course data.

3.3 Results and Discussion

Successes

1. The platform successfully implements a verified instructor-based learning system.

2. Interactive features such as live sessions and discussions enhance the overall learning experience.

3. The system provides a structured and user-friendly environment for both learners and instructors.

4. The database and backend system perform efficiently under typical usage conditions.

Challenges

1. **Real-time Interaction Handling:** Managing live sessions and real-time communication required careful system design and optimization.
2. **Scalability Considerations:** As the number of users grows, additional improvements in database and server infrastructure will be required.
3. **Content Management Complexity:** Ensuring high-quality content and maintaining instructor verification requires continuous monitoring.

Chapter 4

Engineering Standards and Mapping

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

The Elite platform improves access to quality education by providing a structured and reliable digital learning environment. It enables learners from remote or underserved areas to access expert-led courses and develop valuable skills. By ensuring that only verified instructors can create content, the platform enhances trust and learning effectiveness in the digital education space.

4.1.2 Impact on Society & Environment

The Elite system promotes a fully digital learning approach, reducing the need for printed books, notes, and physical learning materials. This paperless environment contributes to environmental sustainability by minimizing resource consumption and supporting eco-friendly education practices.

4.1.3 Ethical Aspects

The system follows ethical standards by ensuring secure handling of user data and maintaining privacy through authentication and authorization mechanisms. Role-based access control ensures that users can only access permitted information, while instructor verification helps maintain content credibility and prevent misinformation.

4.1.4 Sustainability Plan

The Elite Online Learning Platform will ensure sustainability by using scalable cloud services (e.g., Firebase) to handle growing users efficiently. The app will be designed with clean, maintainable code so future updates and features can be added easily. Regular updates and bug fixes will keep the system reliable and secure.

To maintain long-term usage, the platform can introduce free and premium courses, ensuring both accessibility and potential revenue. Additionally, feedback from users will be used to continuously improve the app and keep the content relevant.

4.2 Project Management and Team Work

The development of the Elite platform was carried out through collaborative teamwork, where responsibilities were distributed based on individual expertise:
1. **Frontend Developer:** Designed the user interface and implemented responsive layouts using modern web technologies.
2. **Backend Developer:** Developed server-side logic, APIs, and system functionalities such as authentication and course management.
3. **Database Specialist:** Designed and managed the database structure to ensure efficient data storage and retrieval.

Milestone Tracking

Project progress was managed using a structured timeline to ensure timely completion of each phase:

- **Week 1:** Requirement analysis and system planning
- **Week 2:** UI/UX design and database schema development
- **Week 3:** Implementation of core system modules
- **Week 4:** Testing, debugging, and final system refinement

This structured approach helped maintain consistency and ensured that the project stayed on track.

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

In this section, provide a mapping of the problem and provided solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

PO's	Justification
PO1	Justification of PO1 attainment
PO2	Justification of PO2 attainment
PO3	Justification of PO3 attainment

4.3.2 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 4.2). For P1, you need to put another mapping with

Knowledge profile and rational thereof.

Table 4.2: Mapping with complex problem solving.

EP1 Dept of Knowledge	**EP2** Range of Conflicting Requirements	**EP3** Depth of Analysis	**EP4** Familiarity of Issues	**EP5** Extent of Applicable Codes	**EP6** Extent Of Stakeholder Involvement	**EP7** Inter-dependence
Knowledge of mobile app development, UI/UX design, database management, and basic backend integration.	Balancing user-friendly design with performance, managing data usage vs rich content (videos), and simplicity vs feature richness.	Includes analyzing user needs, app performance, data handling, and efficient content delivery.	Uses common development practices and known technologies like mobile frameworks and cloud services.	Allows standard coding practices, UI design guidelines, and basic data security rules.	Involves students (users), developers, and instructors for feedback and improvement.	Different modules (login, courses, videos, quizzes) depend on each other and must work together smoothly.

4.3.3 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 4.3).

Table 4.3: Mapping with complex engineering activities.

EA1 Range of resources	**EA2** Level of Interaction	**EA3** Innovation	**EA4** Consequences for society and environment	**EA5** Familiarity
Uses mobile development tools, cloud services (Firebase), internet connectivity, and multimedia content like videos.	High interaction between users and the app through learning, quizzes, and progress tracking.	a simple and accessible mobile learning system with interactive features like quizzes and progress monitoring.	romote online education and reduces the need for physical materials, though it depends on internet and device usage.	Based on commonly used technologies and standard app development practices, making it relatively familiar and easy to implement.

Chapter 5

Conclusion

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Summary

The Elite App project successfully addresses the need for a high-quality and structured digital learning platform by introducing a verified instructor-based system. It integrates essential features such as course management, live interactive sessions, role-based access, and collaborative learning tools to provide an effective educational environment.

Key takeaways:

1. **Technical Success:** Successfully developed a functional and well-structured platform with a clear system architecture and reliable performance.
2. **Usability:** Designed an intuitive and user-friendly interface that simplifies navigation and enhances the learning experience for both instructors and learners.
3. **Scalability:** Although initially designed for controlled and moderate-scale usage, the system provides a strong foundation for future expansion and advanced implementations.

5.2 Limitation

Despite its strengths, the Elite platform has several limitations:

1. **Scalability:** The current system architecture may require further optimization and infrastructure upgrades to support a significantly large number of users and courses.
2. **Real-time Interaction:** While live sessions and communication features are implemented, further improvements can be made to enhance real-time performance and responsiveness.
3. **Content Moderation Complexity:** Maintaining strict instructor verification and content quality requires continuous administrative effort.

5.3 Future Work

To further improve the Elite platform and expand its capabilities, several enhancements are proposed:

1. **Database and Infrastructure Upgrade:** Transition to more scalable database systems and cloud-based infrastructure to handle large-scale user activity.

2. **AI-Based Recommendations:** Implement intelligent systems to provide personalized course suggestions based on user interests and learning behavior.

3. **Mobile Application Development:** Develop a dedicated mobile application to improve accessibility and user convenience.

4. **Gamification Features:** Introduce elements such as badges, achievements, and leaderboards to increase user engagement and motivation.

5. **Advanced Analytics:** Provide detailed insights for both learners and instructors to track performance and improve learning outcomes.

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> Official Flask Documentation: https://flask.palletsprojects.com/

2. SQLAlchemy ORM

> SQLAlchemy Official Documentation: https://docs.sqlalchemy.org/

3. SQLite Database

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4. User Authentication

> Flask-Login for Authentication: https://flask-login.readthedocs.io/

> Bcrypt Password Hashing: https://pypi.org/project/bcrypt/

5. HTML/CSS for Frontend Design

> W3Schools HTML Reference: https://www.w3schools.com/html/
> W3Schools CSS Reference: https://www.w3schools.com/css/

6. Flask Security Features

> Flask Security Best Practices: https://flask.palletsprojects.com/en/latest/security/

7. Deployment

> https://pythonanywhere.com